Snapdeal AI Virtual KAM — Practical Assignment Submission

Role: AI First Intern — Snapdeal Seller Experience | **Evaluator:** Sourav Chakraborty

1. Executive Summary & Diagnostic Philosophy

Standard chatbots fail sellers by giving generic advice like 'run discounts' or 'buy ads'. In marketplace e-commerce, advising a seller to drop prices when demand is healthy destroys margins. Our AI Virtual KAM (**Priya Sharma**) takes full business ownership by first analyzing seller telemetry:

Metric	Last Month	This Month	Delta	Forensic Business Insight
Sales	■2,00,000	■1,35,000	**-32.5%**	Direct revenue loss of ■65,000
Orders	400	270	**-32.5%**	130 lost orders (~10 orders per missing SKU)
Cancellation Rate	8%	14%	**+75% spike**	**Alert:** Exceeds 10% threshold; triggers ranking penalty
Average Selling Price	■500	■500	**0.0% (Flat)**	**Healthy:** Pricing & buyer demand are NOT the issue
Out-of-Stock SKUs	5 SKUs	18 SKUs	**+260% surge**	**Smoking Gun:** 13 missing hero listings caused 100% of loss

2. Voice Architecture & Real-Time Pipeline

The assistant uses a zero-dependency, browser-first voice architecture: **Web Speech STT** (SpeechRecognition tuned for Indian English / Hinglish) feeds into a **Deterministic KAM Telemetry Brain** (with optional Gemini 1.5 Flash / GPT-4o adapter), and responds via **SpeechSynthesis TTS** with synchronized visual waveform feedback.

- **Sub-150ms Latency:** Zero cloud roundtrips for speech synthesis allows true real-time conversational pacing.
- **Full Interruptibility:** If the seller speaks or clicks 'Interrupt', the KAM instantly yields the floor.
- **Bi-Directional Telemetry Sync:** When discussing cancellations or stockouts, dashboard cards highlight dynamically.

3. Prescriptive 3-Step Turnaround Plan

Step	Action	Timeline	Business Impact
1	**Priority Restock Top 5 Hero SKUs** Focus supplier on 5 items driving 60% of past sales.	48 Hours	Recovers ~■50,000 in immediate sales.
2	**Enable 5-Unit Buffer in Seller Central** Auto-delist items before stock hits zero.	Immediate	Pulls cancellation rate < 8%, avoiding penalties.
3	**Snapdeal Payday Flash Promo on 32 Active SKUs** Run ■40 promo voucher on in-stock catalog.	This Weekend	Restores cash flow and order velocity immediately.

4. Production Scope & Next Steps

- **Multi-Lingual Voice Stack:** Deepgram Nova-2 + Cartesia Sonic for Hindi, Tamil, Telugu, Marathi, and Punjabi.
- **WMS Webhooks:** Direct REST API connection to Snapdeal WMS to automate supplier restock purchase orders.
- **Proactive Outbound Dialing:** Twilio/Exotel outbound calls when cancellations cross 9% before tier penalty hits.